

Evaluated radial flux: exact finite coefficients and signs

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1 Objects, source, and exact chart

This calculation proves the finite coefficient formula for the actual compact-curl moments' auxiliary-evaluation difference. It retains all old-field derivatives, every ordered new pair, and the signs under physical reflection. The source blowup theorem is attributed to OpenAI; its complete proof is not independently certified here.

The source averaging conventions are (3.7)–(3.9), page 17 of *Finite Time Blowup for Navier–Stokes*; the version and actual reading coverage are recorded in Section 7. These averages act on cylindrical components. On $\mathbb{T}^2 = \mathbb{R}^2/\mathbb{Z}^2$ use normalized Haar measure:

$$\Pi_Y f = \int_{\mathbb{T}^2} f dY, \quad \int_{\mathbb{T}^2} 1 dY = 1, \quad R_Y = I - \Pi_Y, \quad \langle f \rangle_\theta = \frac{1}{2\pi} \int_0^{2\pi} f d\theta.$$

We identify $\Pi_Y f$ with its extension constant in Y . The retained coefficients are smooth on the compact space–time patch. Every new factor and its derivatives have radial support in the actual cutoff interval $K_R \subset (0, \infty)$, which is compact. Old coefficients need only be smooth on this interval. Keep the exact fixed-band chart

$$\begin{aligned} 0 < Q \leq 1, \quad A = \tfrac{1}{2} + h, \quad D = \tfrac{1}{2} - h, \quad \varepsilon = Q^h, \\ r = \sqrt{Q}R, \quad z = Q^D Z, \quad t = 1 - QT. \end{aligned} \tag{1}$$

The constants $A_0, \lambda_0, \mu = \lambda_0$ keep their original values. The clock $\nu = \sqrt{A_0}$ is distinct from physical viscosity $\nu_{\text{NS}} > 0$. No parameter is set to one. The physical auxiliary map retained in the moments proof is

$$\begin{aligned} J_g &= \begin{pmatrix} 3 & 1 \\ 1 & 5 \end{pmatrix}, \quad \Lambda_g = 4 - \sqrt{2}, \quad T_g = 4 + \sqrt{2}, \quad b_g = \sqrt{2} - 1, \\ \mathbf{v}_r &= (1, -b_g), \quad \mathbf{v}_t = (b_g, 1), \quad \rho_g = \frac{\log \Lambda_g}{\log T_g}, \quad \kappa_s = 10^{-5}, \\ d_r &= 2((1+h)\rho_g - h\kappa_s), \quad Y(r, t) = \mathbf{v}_r r^{d_r} + \mathbf{v}_t t \pmod{\mathbb{Z}^2}. \end{aligned} \tag{2}$$

It is independent of θ and z . Consequently its evaluation adds no auxiliary chain term to an axial derivative.

2 Exact angular product and coefficient derivatives

Let Λ and J be the finite retained old-wave and new label sets. Their actual angular carriers n_λ, n_j are nonzero integers. For a cylindrical component $s \in \{r, \theta, z\}$ write

$$\begin{aligned} w_{*,s} &= \sum_{\lambda \in \Lambda} \Re(a_{\lambda,s}(r, z, t, Y) e^{in_\lambda \theta}), \\ v_s &= Q^{-A} \sum_{j \in J} \Re(H_{j,s}(R, Z, T, Y) e^{in_j \theta}), \quad H_{j,s} = b_{j,s} e^{i\psi_j}. \end{aligned} \tag{3}$$

Here w_* is the old oscillatory part. An old angular mean has zero angular product with v , since the integral of each $e^{in_j\theta}$ is zero. All displayed coefficients are independent of θ . The phase is decomposed exactly as

$$\psi_j = k_{\beta_j} m_j \Phi_j - n_j \theta, \quad n_j = k_{\beta_j} m_j p_j.$$

Thus ψ_j is real and independent of θ . Using Φ_j again after extracting the angular carrier would count that carrier twice. The actual curl amplitude is

$$t_j = \chi_j B_j L_j y_j, \quad c_j = \frac{i n_{\Phi,j} \times t_j}{k_{\beta_j} m_j |n_{\Phi,j}|^2}, \quad b_j = t_j + \text{curl}_{*,\text{coeff}} c_j.$$

Every cutoff, auxiliary, cylindrical, and evolving-path derivative in this coefficient curl is retained. Here B_j is the source frame, distinct from the norm $B_{j,e,s}$ below. Define

$$d_{j,s} = \partial_Z b_{j,s} + i(\partial_Z \psi_j) b_{j,s}, \quad G_{j,s} = d_{j,s} e^{i\psi_j} = \partial_Z H_{j,s}. \quad (4)$$

The last equality is the product rule. For $a_{\lambda,s}^Q(R, Z, T, Y) = a_{\lambda,s}(\sqrt{Q}R, Q^D Z, 1 - QT, Y)$ the chain rule gives

$$\partial_Z a_{\lambda,s}^Q = Q^D (\partial_Z a_{\lambda,s})^Q. \quad (5)$$

Old coefficients include their complete old phases and all old scale factors; their derivatives include these factors' full derivatives.

For nonzero integers n, m and complex x, y , set

$$K_{n,m}(x, y) = \frac{1}{2} [\mathbf{1}_{n=m} \Re(xy) + \mathbf{1}_{n=-m} \Re(xy)], \quad \rho_{n,m} = \mathbf{1}_{n=m} + \mathbf{1}_{n=-m}. \quad (6)$$

Then

$$\langle \Re(xe^{in\theta}) \Re(ye^{im\theta}) \rangle_\theta = K_{n,m}(x, y), \quad |K_{n,m}(x, y)| \leq \frac{1}{2} \rho_{n,m} |x| |y|. \quad (7)$$

Proof. Writing each real part as half the sum of a complex exponential and its conjugate gives the four frequencies $n+m, n-m, -n+m, -n-m$. For an integer $k \neq 0$, $(2\pi)^{-1} \int_0^{2\pi} e^{ik\theta} d\theta = (e^{2\pi ik} - 1)/(2\pi ik) = 0$, whereas the integral is one for $k = 0$. The surviving terms are exactly

$$\frac{1}{4} [\mathbf{1}_{n+m=0}(xy + \bar{x}\bar{y}) + \mathbf{1}_{n-m=0}(x\bar{y} + \bar{x}y)],$$

which gives the formula. The inequality follows from $|\Re z| \leq |z|$. Since the carriers are nonzero, $\rho_{n,m}$ is zero or one. $\square$

The kernel is symmetric and real bilinear over real scalars. Conjugation commutes with differentiation in real Z , so its explicit formula proves

$$\partial_Z K_{n,m}(x, y) = K_{n,m}(\partial_Z x, y) + K_{n,m}(x, \partial_Z y). \quad (8)$$

3 Full stress and exact evaluation map

Put

$$\mathcal{S} = \langle w_* \otimes v + v \otimes w_* + v \otimes v \rangle_\theta, \quad \mathcal{S} = \Pi_Y \mathcal{S}, \quad F = Q^{2A} \mathcal{S}^Q.$$

Let $\tau_2 = \theta$ and $\tau_1 = z$. Equations (3)–(7) prove

$$\begin{aligned} F_{z,\tau_e} &= Q^A \sum_{\lambda,j} [K_{n_\lambda, n_j}(a_{\lambda,z}^Q, H_{j,\tau_e}) + K_{n_j, n_\lambda}(H_{j,z}, a_{\lambda,\tau_e}^Q)] \\ &\quad + \sum_{i,j} K_{n_i, n_j}(H_{i,z}, H_{j,\tau_e}). \end{aligned} \quad (9)$$

All sums have the finite ranges $\lambda \in \Lambda$, $i, j \in J$; the last sum includes every ordered pair. Before multiplication by Q^{2A} , the cross and quadratic terms carry Q^{-A} and Q^{-2A} , respectively. This proves their displayed powers. The centered stress in chart units is $R_Y F$. Smoothness and compactness of the fixed torus justify differentiation under its integral, hence $\partial_Z R_Y = R_Y \partial_Z$. Termwise use of (8) gives the complete derivative:

$$\begin{aligned} \partial_Z(R_Y F_{z,\tau_e}) = R_Y \Big\{ & Q^A \sum_{\lambda,j} [K_{n_\lambda,n_j}(\partial_Z a_{\lambda,z}^Q, H_{j,\tau_e}) + K_{n_\lambda,n_j}(a_{\lambda,z}^Q, G_{j,\tau_e}) \\ & + K_{n_j,n_\lambda}(G_{j,z}, a_{\lambda,\tau_e}^Q) + K_{n_j,n_\lambda}(H_{j,z}, \partial_Z a_{\lambda,\tau_e}^Q)] \\ & + \sum_{i,j} [K_{n_i,n_j}(G_{i,z}, H_{j,\tau_e}) + K_{n_i,n_j}(H_{i,z}, G_{j,\tau_e})] \Big\}. \end{aligned} \quad (10)$$

Define the flux moments of these actual fields by

$$M_e = \partial_z \int r^e S_{z,\tau_e} dr, \quad M_e^{\text{ev}} = \partial_z \int r^e \mathcal{S}_{z,\tau_e}(r, z, t, Y(r, t)) dr.$$

The preceding compact-curl moments proof identifies them with the physical residual moments: the radial boundary terms vanish and the time, pressure, and viscosity contributions have zero angular mean. Subtracting the displayed definitions here proves directly

$$M_e^{\text{ev}} - M_e = \partial_z \int r^e (R_Y \mathcal{S}_{z,\tau_e})(r, z, t, Y(r, t)) dr. \quad (11)$$

Compact radial support justifies differentiation under this integral. The chart change, and z -independence of Y , now give exactly

$$\begin{aligned} M_e^{\text{ev}} - M_e &= Q^{\gamma_e} \int_{K_R} R^e (R_Y \partial_Z F_{z,\tau_e})(R, Z, T, Y(\sqrt{Q}R, 1 - QT)) dR, \\ \gamma_e &= \frac{e+1}{2} - 2A - D. \end{aligned} \quad (12)$$

The factors are $Q^{(e+1)/2}$ from radial measure, Q^{-2A} from stress, and Q^{-D} from the axial derivative. Zero auxiliary mean does not cancel evaluation: $f(Y) = \cos(2\pi Y_1)$ has $\Pi_Y f = 0$ but $f(0,0) = 1$. Writing $\mathcal{E}f(r, z, t) = f(r, z, t, Y(r, t))$, the exact operator identity is $\mathcal{E} = \Pi_Y + \mathcal{E}R_Y$, because applying $\mathcal{E}$ to $f = \Pi_Y f + R_Y f$ leaves $\Pi_Y f$ unchanged.

4 The coefficient bound and its required norms

For fixed (Z, T) define

$$N_e(f) = \left(\int_{K_R} R^e \sup_Y |f|^2 dR \right)^{1/2}, \quad C_e^\circ = \int_{K_R} R^e \sup_Y |R_Y \partial_Z F_{z,\tau_e}| dR.$$

Use the actual norms

$$\begin{aligned} A_{\lambda,e,s} &= N_e(a_{\lambda,s}^Q), & A'_{\lambda,e,s} &= N_e(\partial_Z a_{\lambda,s}^Q), \\ B_{j,e,s} &= N_e(b_{j,s}), & D_{j,e,s} &= N_e(d_{j,s}). \end{aligned} \quad (13)$$

Smoothness on $K_R \times \mathbb{T}^2$ proves their finiteness. These norms are L^2 in radius and L^∞ in Y . Earlier auxiliary-averaged L^2 norms do not bound these norms above. The complete finite estimate is

$$\begin{aligned} C_e^\circ &\leq Q^A \sum_{\lambda,j} \rho_{n_\lambda,n_j} [A'_{\lambda,e,z} B_{j,e,\tau_e} + A_{\lambda,e,z} D_{j,e,\tau_e} \\ &\quad + D_{j,e,z} A_{\lambda,e,\tau_e} + B_{j,e,z} A'_{\lambda,e,\tau_e}] \\ &\quad + \sum_{i,j} \rho_{n_i,n_j} [D_{i,e,z} B_{j,e,\tau_e} + B_{i,e,z} D_{j,e,\tau_e}]. \end{aligned} \quad (14)$$

Proof. For continuous f on the torus, $|f(Y) - \Pi_Y f| \leq |f(Y)| + \int |f| dY \leq 2 \sup_Y |f|$. Apply this to (10). The factor two cancels the factor $1/2$ in the kernel bound (7). Since ψ_j is real, $|H_j| = |b_j|$ and $|G_j| = |d_j|$. For each remaining pair, weighted Cauchy–Schwarz gives

$$\int_{K_R} R^e \sup_Y |xy| dR \leq \int_{K_R} R^e (\sup_Y |x|) (\sup_Y |y|) dR \leq N_e(x) N_e(y).$$

Applying this to each of the six product types proves the stated bound. Equation (12) then gives $|M_e^{\text{ev}} - M_e| \leq Q^{\gamma_e} C_e^\circ$. $\square$

For $e = 1$ the cross part combines to $2Q^A \sum_{\lambda,j} \rho_{n_\lambda, n_j} (A'_{\lambda,1,z} B_{j,1,z} + A_{\lambda,1,z} D_{j,1,z})$; the ordered quadratic sum is retained. The centering operator's norm is exactly two. For its lower bound take a smooth torus bump $0 \leq \eta_\delta \leq 1$ with maximum one and support measure at most δ . Then $f = -1 + 2\eta_\delta$ has norm one, and at its maximum $|f - \Pi_Y f| = 2(1 - \int \eta_\delta) \geq 2(1 - \delta)$. Such bumps are obtained by rescaling a smooth compact bump in a torus coordinate ball. Letting its support shrink proves the lower bound two; the preceding triangle inequality proves the upper bound.

5 Physical orders and every axial derivative

On $K_r = \sqrt{Q} K_R$ define

$$P_{\lambda,e,s}^2 = \int_{K_r} r^e \sup_Y |a_{\lambda,s}|^2 dr, \quad \dot{P}_{\lambda,e,s}^2 = \int_{K_r} r^e \sup_Y |\partial_z a_{\lambda,s}|^2 dr.$$

Changing radius and using (5) proves

$$A_{\lambda,e,s} = Q^{-(e+1)/4} P_{\lambda,e,s}, \quad A'_{\lambda,e,s} = Q^{D-(e+1)/4} \dot{P}_{\lambda,e,s}. \quad (15)$$

For the retained old orders write the exact factorizations $P_{\lambda,e,s} = Q^{\omega_{\lambda,e,s}} W_{\lambda,e,s}$ and $\dot{P}_{\lambda,e,s} = Q^{\dot{\omega}_{\lambda,e,s}} \dot{W}_{\lambda,e,s}$. At fixed $Q > 0$ these define $W, \dot{W}$ from the actual norms. No uniformity in Q is inferred. With $\alpha_e = (e+1)/4 - A$, multiplication of (14) by Q^{γ_e} gives the two old-factor routes with their original component indices:

$$Q^{\alpha_e + \dot{\omega}_{\lambda,e,s}} \dot{W}_{\lambda,e,s} B_{j,e,k}, \quad Q^{\alpha_e + \omega_{\lambda,e,s} - D} W_{\lambda,e,s} D_{j,e,k}. \quad (16)$$

Their exponents follow respectively from $\gamma_e + A + D - (e+1)/4 = \alpha_e$ and $\gamma_e + A - (e+1)/4 = \alpha_e - D$. No old field order is identified with its derivative order. The quadratic factor is

$$Q^{\gamma_e} = Q^{2\alpha_e - D}, \quad \gamma_2 = -h, \quad \gamma_1 = -\frac{1}{2} - h. \quad (17)$$

Substitution of A, D proves both exponents. For the source's $h > 0$ these explicit bound factors grow as Q decreases. This proves neither growth nor a sign for the actual moment: the coefficient norms can still depend on Q .

To retain every axial derivative, define

$$b_j^{[0]} = b_j, \quad b_j^{[m+1]} = \partial_Z b_j^{[m]} + i(\partial_Z \psi_j) b_j^{[m]}, \quad B_{j,e,s}^{[m]} = N_e(b_{j,s}^{[m]}), \quad A_{\lambda,e,s}^{[m]} = N_e(\partial_Z^m a_{\lambda,s}^Q).$$

Induction by the product rule proves $\partial_Z^m H_j = e^{i\psi_j} b_j^{[m]}$. The recurrence includes all derivatives of $\partial_Z \psi_j$. Differentiating (8) repeatedly and combining adjacent binomial coefficients proves

$$\partial_Z^q K_{n,m}(x, y) = \sum_{\ell=0}^q \binom{q}{\ell} K_{n,m}(\partial_Z^\ell x, \partial_Z^{q-\ell} y).$$

For $k \geq 0$ an integer, let $C_{e,k}^\circ = \int_{K_R} R^e \sup_Y |R_Y \partial_Z^{k+1} F_{z,\tau_e}| dR$. The proof of (14), term by term, now yields

$$\begin{aligned} C_{e,k}^\circ &\leq Q^A \sum_{\lambda,j} \rho_{n_\lambda,n_j} \sum_{\ell=0}^{k+1} \binom{k+1}{\ell} [A_{\lambda,e,z}^{[\ell]} B_{j,e,\tau_e}^{[k+1-\ell]} + B_{j,e,z}^{[\ell]} A_{\lambda,e,\tau_e}^{[k+1-\ell]}] \\ &\quad + \sum_{i,j} \rho_{n_i,n_j} \sum_{\ell=0}^{k+1} \binom{k+1}{\ell} B_{i,e,z}^{[\ell]} B_{j,e,\tau_e}^{[k+1-\ell]}. \end{aligned} \quad (18)$$

Apply ∂_z^k to (12). Since Q is fixed and Y independent of z , each derivative adds $Q^{-D} \partial_Z$. Therefore

$$|\partial_z^k (M_e^{\text{ev}} - M_e)| \leq Q^{(e+1)/2-2A-(k+1)D} C_{e,k}^\circ. \quad (19)$$

At every finite derivative order the same compact patch provides an integrable radial support, so all interchanges of derivative and integral are justified. No infinite-order bound is asserted.

6 The signs changed by physical reflection

Let $H = \text{diag}(1, -1, 1)$ and transform both complete velocity fields by $u^H(x, t) = Hu(Hx, t)$. Set $f^H(x, t) = Hf(Hx, t)$ and $p^H(x, t) = p(Hx, t)$. Orthogonality and the chain rule give

$$\begin{aligned} \partial_t u^H &= H(\partial_t u)(Hx, t), \quad (u^H \cdot \nabla) u^H = H((u \cdot \nabla) u)(Hx, t), \\ \Delta u^H &= H(\Delta u)(Hx, t), \quad \nabla p^H = H(\nabla p)(Hx, t), \quad \nabla \cdot u^H = (\nabla \cdot u)(Hx, t). \end{aligned}$$

Indeed $\partial_i u_a^H = H_{ab} H_{ci} \partial_c u_b$; contraction using $\sum_i H_{ci} H_{di} = \delta_{cd}$ proves the convection and Laplacian identities. Taking the trace of $H(\nabla u)H$ proves the divergence identity. Thus the full Navier–Stokes residual transforms by H at the same positive viscosity. The map is its own inverse. The change of variables $x = Hy$ has absolute Jacobian one and preserves the L^2 and L^∞ norms, smoothness, and compactness of support. It transports the source witness with its transformed force.

In right-handed cylindrical components Hx has coordinates $(r, -\theta, z)$, and

$$(u_r^H, u_\theta^H, u_z^H)(r, \theta, z, t) = (u_r, -u_\theta, u_z)(r, -\theta, z, t).$$

To verify this, use $He_r(-\theta) = e_r(\theta)$, $He_\theta(-\theta) = -e_\theta(\theta)$, and $He_z = e_z$. Applying this rule to both w_* and v negates every $z\theta$ product and preserves every zz product. Periodicity and $\vartheta = -\theta$ in the angular integral then prove

$$\mathcal{S}_{z,\theta}^H = -\mathcal{S}_{z,\theta}, \quad \mathcal{S}_{z,z}^H = \mathcal{S}_{z,z}.$$

Reflection preserves r, z, t , and hence the phase map (2). Centering, evaluation, radial integration, and ∂_z preserve these component signs. Consequently

$$\begin{aligned} (M_2^H, (M_2^{\text{ev}})^H) &= -(M_2, M_2^{\text{ev}}), \\ (M_1^H, (M_1^{\text{ev}})^H) &= (M_1, M_1^{\text{ev}}). \end{aligned} \quad (20)$$

The evaluation differences have the same respective signs, and their absolute envelopes are unchanged. On coefficient arrays the map is $n \mapsto -n$, $(a_r, a_\theta, a_z) \mapsto (a_r, -a_\theta, a_z)$, and the same component rule for b_j, d_j , with ψ_j unchanged. Directly from (6), $K_{-n,-m}(sx, ty) = stK_{n,m}(x, y)$ for $s, t \in \{1, -1\}$; both indicators are unchanged. This also proves (20) from the full coefficient expansion.

A positive scale does not require positive flux: $K_{n,n}(1, -1) = -1/2$. Reflection does not negate the entire flux: it preserves the axial moment. The viscosity remains the positive ν_{NS} in the full increment

$$\partial_t v + (u_* \cdot \nabla) v + (v \cdot \nabla) u_* + (v \cdot \nabla) v - \nu_{\text{NS}} \Delta v + \nabla \pi.$$

Its zero averaged moment does not remove it from this PDE.

7 Source and proof record

The source is OpenAI, *Finite Time Blowup for Navier–Stokes* (2026). Its official announcement, dated 8 September 2026, describes finite-time blowup from zero velocity under smooth forcing, with finite energy. The kernel, finite bounds, exponents, and reflection identities in this addendum are proved in full above.

The TeX inspected for the averaging notation is an independent reconstruction at commit `5e162f34`, specifically `reconstruction/sections/pp013-018.tex`, lines 284–374, including (3.7)–(3.9). It is not original author TeX. The supplied bounded search did not locate original author TeX. The local record preserves the full version, fetched-file hashes, and actual reading coverage. The related archive is DOI 10.5281/zenodo.22852310. No personal attribution is inferred from an account name.

The cumulative reader retains the complete earlier physical compact-curl moment derivation and phase map. The present addition fills its finite evaluated-flux coefficient calculation. Uniform bounds for an increasing family of bands and the distinct modified sequence’s terminal smooth-force extension do not follow from the finite norms proved here.

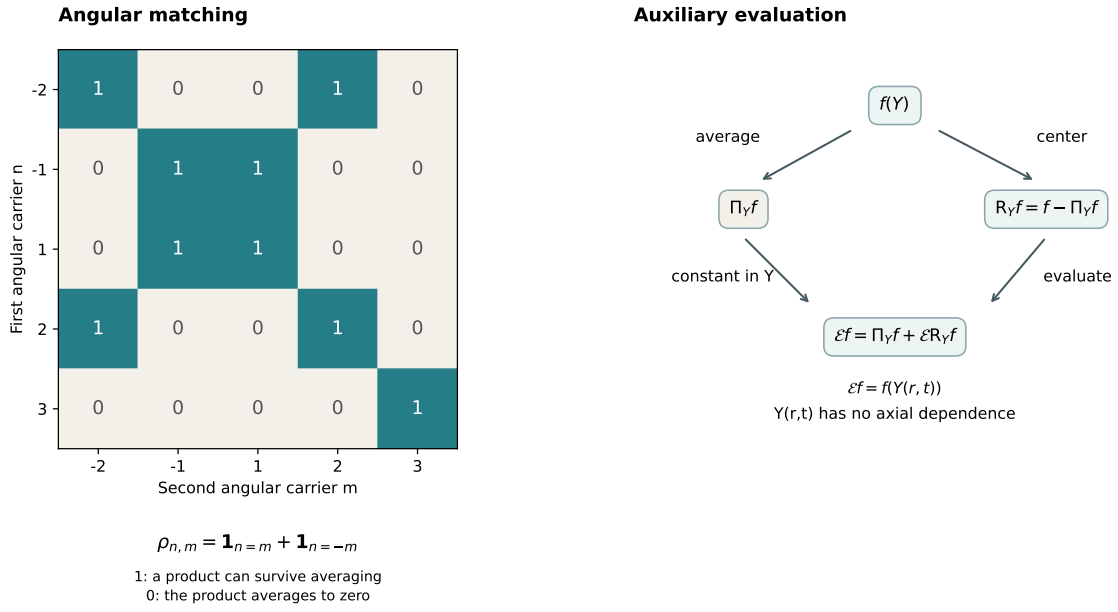


Figure 1: Exact angular matching and auxiliary evaluation. Left: the displayed finite carriers illustrate $\rho_{n,m}$; a value one permits a signed kernel, not necessarily positive flux. Right: evaluation retains the centered auxiliary part. Proofs: (6)–(7) and (11)–(12). Source conventions: (3.7)–(3.9). This is a diagram of exact maps, not a sampled PDE solution. Reproducible source: `draw_flux_maps.py`.